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Applied Time Series Analysis
STA457H5S, Winter 2021
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Test 2 Solutions

Question 2 (25pts). Show that the value at lag 2 of the partial ACF of the MA(1) process

$$X_t = W_t + \theta W_{t-1}$$

where $\{W_t\} \sim wn(0, \sigma_w^2)$, is

$$\phi_{22} = \frac{-\theta^2}{1 + \theta^2 + \theta^4}$$

Solution. The partial ACF at lag h , denoted by ϕ_{hh} , comes from the best linear predictor of X_{h+1} , which has the following form

$$\hat{X}_{h+1} = \sum_{i=1}^h \phi_{hi} X_{h+1-i}$$

In particular, we have

$$\hat{X}_3 = \phi_{21} X_2 + \phi_{22} X_1$$

Also, the best linear predictor satisfies the following equations:

$$\begin{cases} Cov(X_3 - \hat{X}_3, X_1) = 0 \\ Cov(X_3 - \hat{X}_3, X_2) = 0 \end{cases}$$

This, in turn, gives us

$$\begin{aligned} Cov(X_3 - \hat{X}_3, X_1) &= Cov(X_3 - \phi_{21} X_2 - \phi_{22} X_1, X_1) \\ &= Cov(X_3, X_1) - \phi_{21} Cov(X_2, X_1) - \phi_{22} Cov(X_1, X_1) \\ &= \gamma_2 - \phi_{21} \gamma_1 - \phi_{22} \gamma_0 = 0 \end{aligned}$$

and

$$\begin{aligned}
 Cov(X_3 - \hat{X}_3, X_2) &= Cov(X_3 - \phi_{21}X_2 - \phi_{22}X_1, X_2) \\
 &= Cov(X_3, X_2) - \phi_{21}Cov(X_2, X_2) - \phi_{22}Cov(X_1, X_2) \\
 &= \gamma_1 - \phi_{21}\gamma_0 - \phi_{22}\gamma_1 = 0
 \end{aligned}$$

Next, recall that the autocovariance function of an MA(1) process is given as follows:

$$\gamma_h = \begin{cases} \sigma_w^2(1 + \theta^2), & \text{if } h = 0 \\ \sigma_w^2\theta, & \text{if } |h| = 1 \\ 0, & \text{otherwise} \end{cases}$$

Thus, we have to solve the equations

$$\begin{cases} \phi_{21}\gamma_1 + \phi_{22}\gamma_0 = 0 \\ (1 - \phi_{22})\gamma_1 - \phi_{21}\gamma_0 = 0 \end{cases}$$

which yields

$$\phi_{22} = \frac{-\theta^2}{1 + \theta^2 + \theta^4}$$

Question 3 (25pts). Identify the following model as ARMA(p,q) process, and determine whether it is causal and/or invertible. Here, we have $\{W_t\} \sim wn(0, 1)$. If the model is shown to be causal, then further compute its corresponding first five coefficients $\psi_0, \psi_1, \psi_2, \psi_3, \psi_4$ in the causal linear process representation $X_t = \sum_{j=0}^{\infty} \psi_j W_{t-j}$.

$$X_t - X_{t-1} = W_t - \frac{1}{2}W_{t-1} - \frac{1}{2}W_{t-2}$$

Solution. The first step, as always, is to check for parameter redundancy. To this end, we find the roots of the AR and MA characteristic polynomials, i.e.,

$$\phi(B) = 1 - B = 0 \implies B = 1$$

$$\theta(B) = 1 - \frac{1}{2}B - \frac{1}{2}B^2 = 0 \implies B_1 = 1, \quad B_2 = -2$$

It is seen that both polynomials share a common root. Thus, we need to eliminate the factor corresponding to the root $B = 1$ from both polynomials, i.e., we need to remove the factor $(1 - B)$ in order to derive the following actual (reduced) forms of the AR and MA polynomials:

$$\phi_{\text{red}}(B) = 1$$

$$\theta_{\text{red}}(B) = 1 + \frac{1}{2}B$$

Then, we can specify the given time series to be an ARMA(0,1) process (or, in other words, an MA(1) process). Hence, this model is invertible as $\theta_{\text{red}}(B) = 0$ has root -2 . Also, note that the irredundant representation of the AR polynomial, $\phi_{\text{red}}(B) = 1$, has no roots. So, the process will become a causal one. To elaborate more on the reason why the case of *no roots* implies causality, one might consider the following reasons:

1- To have a causal model, all we want is to be able to write X_t as a function of W_t (white noise). In our obtained MA(1) model, this condition is already satisfied since we can write the irredundant model as follows:

$$X_t = W_t + \frac{1}{2}W_{t-1}$$

2- Another way to validate this result is to note that the following two statements

1. A process is causal if all roots of the AR polynomial lie outside the unit circle
2. A process is causal if none of the roots of the AR polynomial fall inside the unit circle

are equivalent. Since the first statement does not provide any explicit instruction on what to do in case of *no roots*, we can resort to the applications of the second statement. Since $\phi_{\text{red}}(B) = 1$ has no roots (it's a constant function parallel to the horizontal axis which does not intersect it), it does not violate the conditions of the second statement, which in turn makes the process causal.

Next, the power series ψ is given by the expansion of

$$\psi(B) = \frac{\theta_{\text{red}}(B)}{\phi_{\text{red}}(B)} = \frac{1 + \frac{1}{2}B}{1} = 1 + \frac{1}{2}B + 0B^2 + 0B^3 + 0B^4$$

which implies that

$$\psi_0 = 1, \psi_1 = \frac{1}{2}, \psi_2 = 0, \psi_3 = 0, \psi_4 = 0$$